50.007 Machine Learning

Lecture 6

K-Medoids Clustering

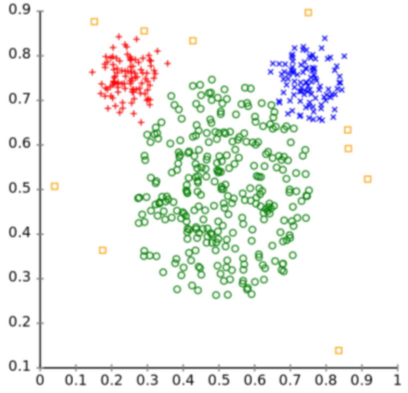
What is clustering

- Form of *unsupervised* learning - no information from teacher
- The process of partitioning a set of data into a set of meaningful (hopefully) sub-classes, called *clusters*

Cluster:

- collection of data points that are "similar" to one another and collectively should be treated as group
- as a collection, are sufficiently different from other groups

What is clustering



Clustering Problem.

Input.

Training data $\mathcal{S}_n = \{x^{(i)}; i = 1, 2, \ldots, n\}$, each $x^{(i)} \in \mathbb{R}^d$. Integer k.

Output.

Clusters $\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_k \subset \{1, 2, \ldots, n\}$ such that every data point is in one and only one cluster.

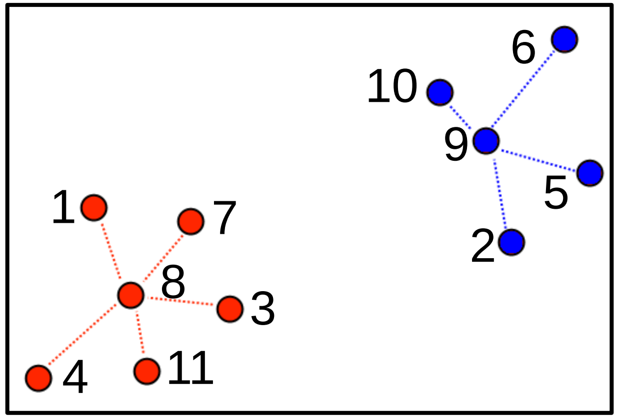
Some clusters could be empty!

How to Specify a Cluster

- By listing all its **elements**

$$\mathcal{C}_1 = \{1,3,4,7,8,11\}$$

$$\mathcal{C}_2 = \{2,5,6,9,10\}$$

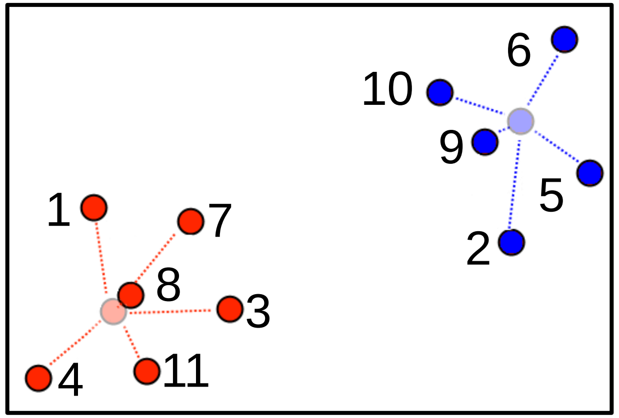


How to Specify a Cluster

- Using a **representative**
  a. A point in center of cluster (centroid)
  b. A point in the training data (exemplar)

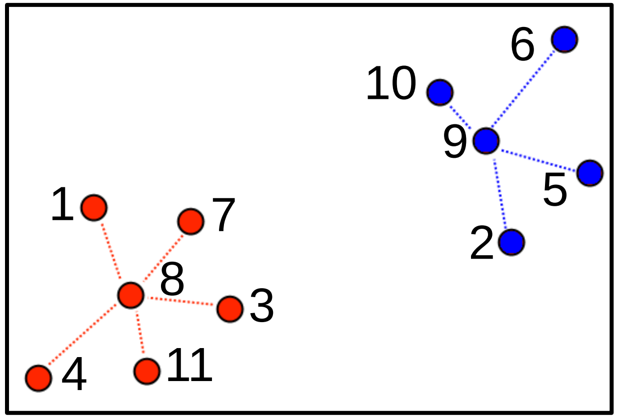
Each point $x^{(i)}$ will be assigned the closest representative.

$z^{(1)} = \binom{1}{1}, z^{(2)} = \binom{5}{4}$



centroid

$z^{(1)} = 8, z^{(2)} = 9$



exemplar

K-Means Algorithm

1. Initialize centroids $z^{(1)}, \dots, z^{(k)}$ from the data.
2. Repeat until no further change in training loss:

   a. For each $j \in \{1, \dots, k\}$,
   $$\mathcal{C}_j = \{\, i \text{ such that } x^{(i)} \text{ is closest to } z^{(j)} \,\}.$$

   b. For each $j \in \{1, \dots, k\}$,
   $$z^{(j)} = \frac{1}{|\mathcal{C}_j|} \sum_{i \in \mathcal{C}_j} x^{(i)} \quad \text{(cluster mean)}$$

Animation: https://towardsdatascience.com/k-means-clustering-introduction-to-machine-learning-algorithms-c96bf0d5d57a

K-Medoids

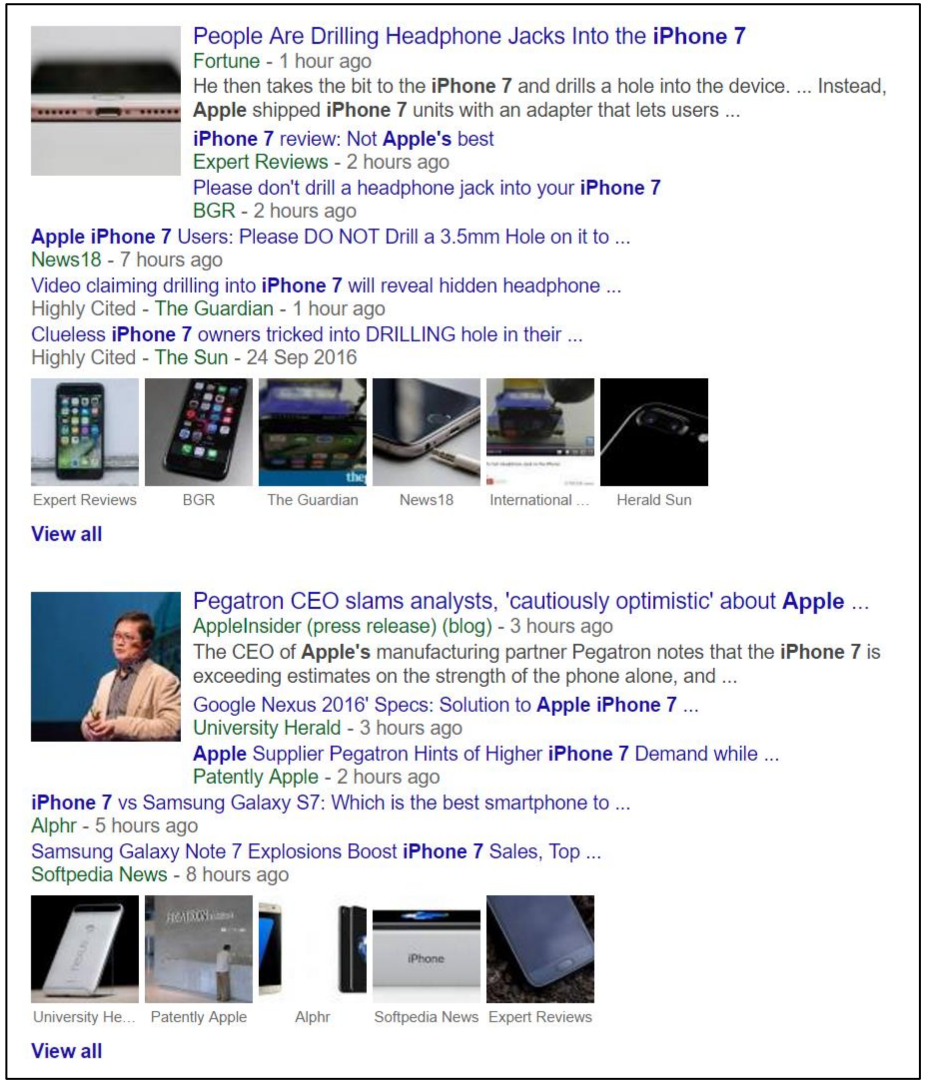
used to find similarity in content for related news articles.

Use exemplars
instead of centroids.

e.g. Google News.

Repeat until convergence:

- Find best clusters given exemplars
- Find best exemplars given clusters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K-Medoids Algorithm

1. Initialize exemplars $z^{(1)}, \dots, z^{(k)} \subseteq \{x^{(1)}, \dots x^{(n)}\}$
2. Repeat until no further change in training loss:

   a. For each $j \in \{1, \dots, k\}$,
      $\mathcal{C}^{\mathrm{j}} = \{\, i$ such that $x^{(i)}$ is closest to $z^{(j)}\,\}$.

   b. For each $j \in \{1, \dots, k\}$, set $z^{(j)}$ to be the point in $C^{(j)}$ that minimizes $\sum_{i \in \mathcal{C}^{\mathrm{j}}} d(x^{(i)}, z^{(j)})$ $\longrightarrow$

different from k-means
⇒ you are finding the exemplars in the clusters.

* It depends on the size of the cluster, since the algorithm has to loop through the datapoints iteratively to find the better exemplar in the cluster.

K-Medoids Algorithm

looping through the cluster?

yes looping through & swapping if the cost is lesser.

1. Initialize exemplars $z^{(1)}, \ldots, z^{(k)} \subseteq \{x^{(1)}, \ldots x^{(n)}\}$
2. Repeat until no further change in training loss:

   a. For each $j \in \{1, \ldots, k\}$,
   $\mathcal{C}^{\mathrm{j}} = \{\, i$ such that $x^{(i)}$ is closest to $z^{(j)}$ $\}$.

   b. For each $j \in \{1, \ldots, k\}$,
   set $z^{(j)}$ to be the point in $C^{(j)}$ that minimizes $\sum_{i \in \mathcal{C}^{\mathrm{j}}} d(x^{(i)}, z^{(j)})$

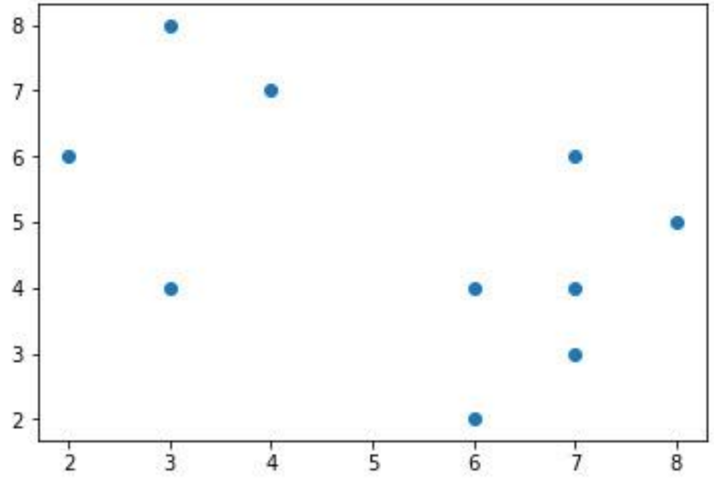
For each data point, $x^{(i)}$ which is not a medoid:

1. Swap $z^{(j)}$ and $x^{(i)}$, associate each data point to the swapped medoid, recompute the cost.
2. If the total cost is more than that in the previous step, undo the swap.

Example: K-Medoids

Example: K-Medoids

- Consider the following set of points



$x^{(1)}$	2	6
$x^{(2)}$	3	4
$x^{(3)}$	3	8
$x^{(4)}$	4	7
$x^{(5)}$	6	2
$x^{(6)}$	6	4
$x^{(7)}$	7	3
$x^{(8)}$	7	4
$x^{(9)}$	8	5
$x^{(10)}$	7	6

- We will consider L1 distance $d(x^{(i)}, z^{(j)}) = |x^{(i)} - z^{(j)}|$

Source: https://en.wikipedia.org/wiki/K-medoids

Example: K-Medoids

- Let the randomly selected 2 medoids be

$$z^{(1)} = (3,4) = x^{(2)}$$
$$z^{(2)} = (7,4) = x^{(8)}$$

- The cost of each non-medoid point with the medoids is calculated and tabulated:

Data object		Distance to	
i	$x^{(i)}$	$z^{(1)} = (3,4)$	$z^{(2)} = (7,4)$
1	(2, 6)	3	7
2	(3, 4)	0	4
3	(3, 8)	4	8
4	(4, 7)	4	6
5	(6, 2)	5	3
6	(6, 4)	3	1
7	(7, 3)	5	1
8	(7, 4)	4	0
9	(8, 5)	6	2
10	(7, 6)	6	2
Cost			

* Take the minimum & assign the cluster

Example: K-Medoids

- Let the randomly selected 2 medoids be

$$z^{(1)} = (3,4)$$
$$z^{(2)} = (7,4)$$

- The total cost of this clustering is:

Cluster 1: (3+0+4+4) = 11

Cluster 2: (3+1+1+0+2+2) = 9

Total: 20

Data object		Distance to	
i	$x^{(i)}$	$z^{(1)} = (3,4)$	$z^{(2)} = (7,4)$
1	(2, 6)	**3**	7
2	(3, 4)	**0**	4
3	(3, 8)	**4**	8
4	(4, 7)	**4**	6
5	(6, 2)	5	**3**
6	(6, 4)	3	**1**
7	(7, 3)	5	**1**
8	(7, 4)	4	**0**
9	(8, 5)	6	**2**
10	(7, 6)	6	**2**
Cost		**11**	**9**

Example: K-Medoids

- Updating Z_2 with a non-medoid point, O'

$$z^{(1)} = (3,4)$$
$$O' = (7,3)$$

- The total cost of this clustering is:

Cluster 1: (3+0+4+4) = 11

Cluster 2: (2+2+0+1+3+3) = 11

Total: 22

i	$z^{(1)}$		$x^{(i)}$		dist
1	3	4	2	6	**3**
3	3	4	3	8	**4**
4	3	4	4	7	**4**
5	3	4	6	2	5
6	3	4	6	4	3
8	3	4	7	4	4
9	3	4	8	5	6
10	3	4	7	6	6

i	O'		$x^{(i)}$		dist
1	7	3	2	6	8
3	7	3	3	8	9
4	7	3	4	7	7
5	7	3	6	2	**2**
6	7	3	6	4	**2**
8	7	3	7	4	**1**
9	7	3	8	5	**3**
10	7	3	7	6	**3**

How do we find the first 2 exemplar to initialise?

⇒ If not graphically representative, take average of datapoints & get the closest data point as exemplar.

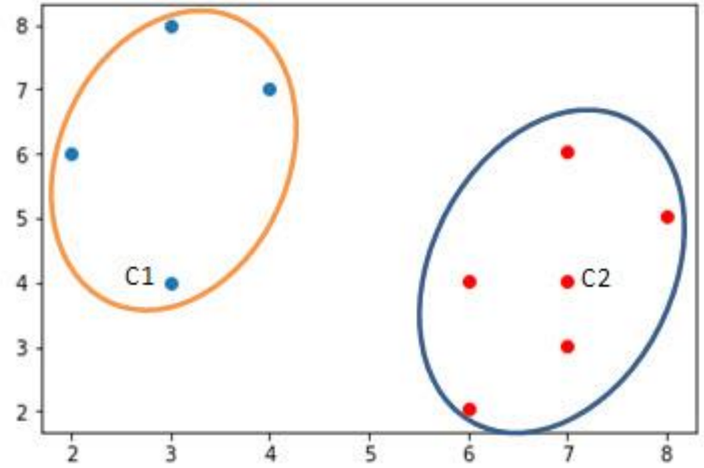
Example: K-Medoids

- The total cost of this clustering is:

Cluster 1: (3+0+4+4) = 11

Cluster 2: (2+2+0+1+3+3) = 11

Total: 22 > 20 → No swapping



↳ loop through all datapoints to find exemplar which has minimum loss.

Discussion

Initialization

Optimization

How to determine the # of clusters? => k-value.
How to initialise better?

- Empty clusters
  - Pick data points to initialize clusters
- Bad local minima
  - Initialize many times and pick solution with smallest training loss
  - Pick good starting positions

Initialization

Optimization

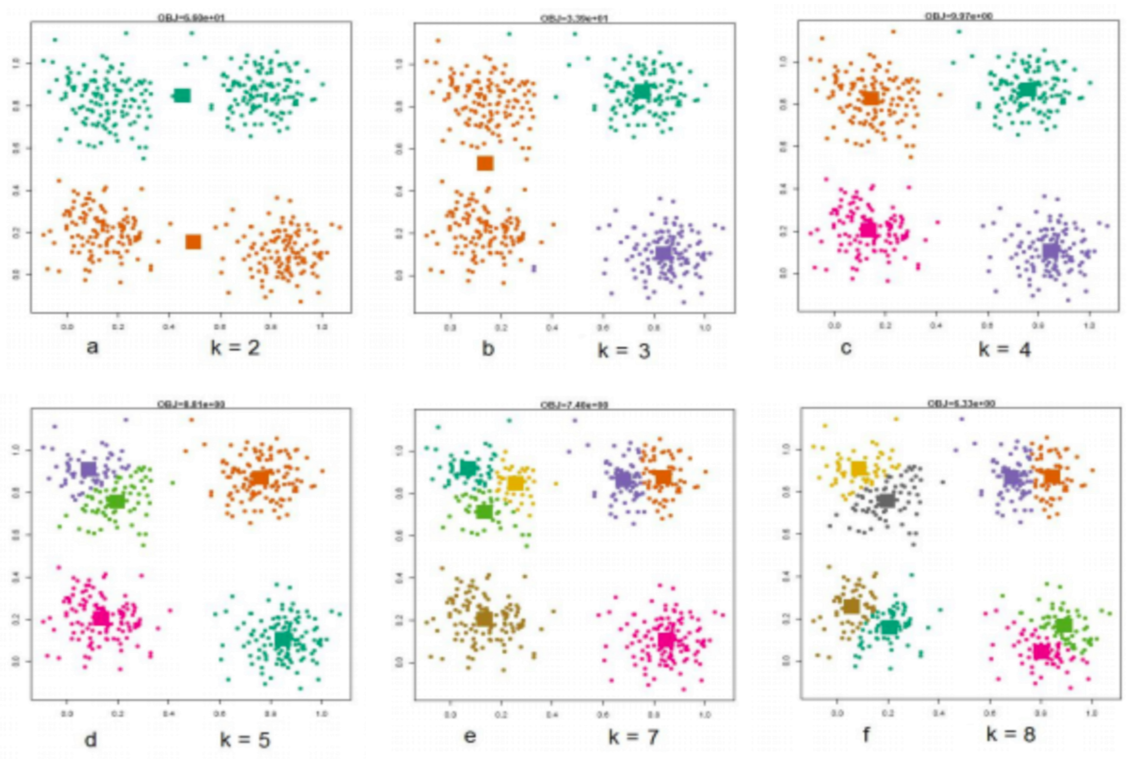
Starting position of centroids

Final position of centroids

Problem. How to choose good starting positions?
Solution. Place them far apart with high probability.

Number of Clusters

Generalization



Number of Clusters

Generalization

How do we choose k, the optimal number of clusters?

- Elbow method
  - Training Loss
  - Validation Loss

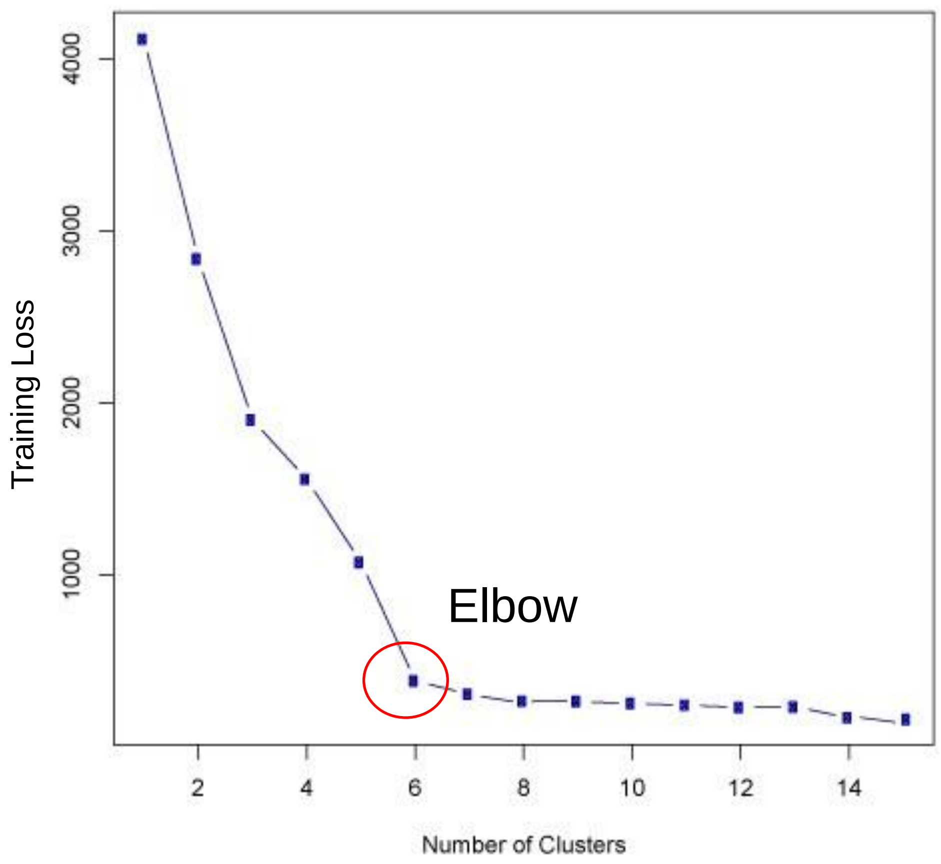
By changing the k-value = # of clusters & finding the training loss, we can find a point where the k-value is meaningful & minimum training loss (see next slide)

- Semi-supervised learning
  - Accuracy in supervised task

↳ find the # of meaning features in unsupervised learning & use it to determine the value of K in supervised learning since the meaningful features will highly likely be useful in supervised learning as well.

Elbow Method

Generalization



Summary

- The **K-medoids algorithm** shares the properties of K-means that we discussed (each iteration **decreases the cost**; the algorithm always **converges**; different starts gives different final answers; it **does not achieve the global minimum**)

- **K-medoids is computationally harder** than K-means (because of step 2: computing the medoid is harder than computing the average)

- Remember, **K-medoids** has the (potentially important) property that the **centers are located among the data points themselves**

Intended Learning Outcomes

- What is k-Medoids algorithm and how is it different from k-means.
- Why the initialization of the k-means algorithm matters. $\because$ of concept of global & local minimization loss.
- How to determine the k used in the k-means algorithm.